

Xuanhe, Yang

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EDUCATION

Nanyang Technological University, Singapore

08/2026 - Present

- Master of Science in Artificial Intelligence(Expected)

Tongji University, China

09/2022 - 06/2026

- Bachelor of Engineering in Software Engineering(Machine Intelligence track)

CGPA:89.82/100

- Core modules: Machine Learning, Swarm Intelligence, Principles of Databases and Applications, Curriculum Design for Database, Computer Network, Computer Architecture, High-Level Language Programming , Data Structures, Computer Organization, Programming Paradigms, Algorithms Design and Analysis, User Interface Interaction, Operating Systems, Introduction to Artificial Intelligence, Computer Vision, System Analysis and Design, Software Design Patterns, Software Engineering, Automatic Speech Recognition

RESEARCH INTERESTS

- Multi-Agent Systems, Reinforcement Learning, Reliable Machine Learning, Large Language Models

PATENT

- **Xuanhe Yang**, Linyang Bian ,Zhenghai Chen , Ruiyuan Duan , Neural Network-based Equipment Lifespan Prediction Web Application and System Construction Method. - Patent Application No.202411947358.5, published in China National Intellectual Property Administration (CNIPA) Patent Gazette, Vol.41, No.1702, on April 25, 2025.

RESEARCH EXPERIENCE

- **Multi-UAV/Multi-Truck Air-Ground Collaborative Last-Mile Delivery System** 02/2026 - 06/2026

Undergraduate Thesis, School of Software Engineering, Tongji University (Advisor: Prof. Qingjiang Shi)

Aim: Designing a scalable, hierarchical multi-agent decision-making framework for large-scale heterogeneous fleet coordination (multi-UAV + multi-truck) in last-mile delivery, combining combinatorial task allocation with sequential multi-agent reinforcement learning

- **Designed a three-tier hierarchical optimization framework** decomposing the large-scale coordination problem into region-level clustering (K-Means++), auction-based task allocation, and multi-agent reinforcement learning (IPPO) for real-time cooperative decision-making among heterogeneous agents.
- **Formulated task allocation as a combinatorial auction problem** to assign delivery tasks across a heterogeneous fleet under capacity and route constraints, mirroring auction-based and DCOP-style formulations used for large-scale multi-agent coordination and mobility-on-demand systems.
- **Trained cooperative UAV-truck policies via IPPO** in a custom PettingZoo simulation environment, validating that the hierarchical decomposition (allocation + learning) scales to instance sizes intractable for centralized planning.
- **Demonstrated that decoupling combinatorial task assignment** from sequential decision-making yields a tractable and extensible framework for real-world, large-scale collective multi-agent systems — directly relevant to auction mechanism design, DCOP, and collective multi-agent planning for city-scale logistics and mobility-on-demand applications.

- **Motion Data Quality Assessment and Diversity Evaluation System**

09/2025 - 01/2026

Research Assistant, Prof. Peng Yin's Lab, City University of Hong Kong

Aim: Building a universal hand motion dataset for robotics (analogous to ImageNet for computer vision) with intelligent quality control and diversity evaluation frameworks

- **Investigated reliability risks in large-scale robotic manipulation data**, developing an LLM-driven quality assessment pipeline to detect corrupted, low-quality, or out-of-distribution samples in hand motion

trajectories — directly analogous to OOD detection in the visual/language domains, but applied to a new data modality (kinematic + tactile).

- **Designed diversity evaluation metrics to characterize distributional coverage of the dataset** (gesture variety, motion complexity, task coverage), effectively formalizing a notion of "in-distribution support" for robotic manipulation data — a question closely related to characterizing Pin vs. unseen/underrepresented regions of the data distribution.
- **Explored multi-modal fusion** (tactile sensing, vision, kinematics) to study how representation choice affects the identifiability of unreliable or anomalous samples, motivating current work on cross-domain generalization under distribution shift.

Github: https://github.com/CyberOrigin2077/open_cyber_glove

- **MaintainMind: Aircraft Engine Intelligent Maintenance Web Application** 02/2025 - 04/2025

Project Software Part Leader, Research Assistant in Prof. Chunyan Duan's Lab, Tongji University

Aim: Developing an intelligent predictive maintenance web application for aircraft engines using advanced deep learning technologies

- **Won First Prize** in Tongji University "Excellence Cup" and "Challenge Cup" National Competition, demonstrating exceptional innovation in aerospace maintenance technology
- **Developed multi-model RUL prediction framework** integrating CNN-LSTM, CNN-Transformer models for precise remaining useful life forecasting of aircraft engines
- **Created comprehensive maintenance management system** featuring equipment center, report management, data simulation, and alert recording modules for complete lifecycle equipment management

Github: <https://github.com/Augensternst/MaintainMind-Aircraft-Engine-PHM>

- **Neural Network-Based Equipment Life Prediction Web Application System (Patent)** 10/2024 - 01/2025

Student First Author, Supervisor: Prof. Chunyan Duan, Tongji University

Aim: Developing an intelligent web-based system for industrial equipment remaining useful life (RUL) prediction using neural networks

- **Led the system architecture design** with distributed platform including data collection, processing, prediction model, application service, and user interaction layersMade users as audience members, evaluating agents' statements in real-time and conducting deeper post-discussion interaction
- **Implemented deep learning models** including Artificial Neural Networks (ANN) and Long Short-Term Memory (LSTM) networks for equipment degradation pattern recognition and lifespan forecastingExplored
- **Built full-stack web application** using Vue.js frontend with Element-plus components, Express.js backend with RESTful APIs, MySQL database, and Redis caching for real-time equipment monitoring
- **Created intelligent maintenance decision support** generating automated reports with remaining lifespan, health status, risk levels, and maintenance recommendations

Github:<https://github.com/Augensternst/A-Web-Application-System-for-Equipment-Life-Prediction-Based-on-Neural-Networks>

- **Smart Community Property Management System** 09/2024 - 12/2024

Research Assistant in Prof. Weichao Chen's Lab, Tongji University

Aim: Developing a comprehensive digital platform for intelligent community property services and maintenance management

- **Collaborated in comprehensive system development** including online repair reporting, staff rating system, AI assistant, and video retrieval functionalities for modern residential community management
- **Participated in full-stack development** using WeChat Mini Program for mobile frontend, web application for admin interface, MySQL database for data persistence, and RESTful API architecture
- **Contributed to multi-modal AI integration** including CLIP-based video search system for maintenance-related footage retrieval and large language model integration for intelligent chatbot assistance

Github: <https://github.com/Augensternst/Smart-Community-Property-Management>

- **Cross-Modal Web Video Retrieval System for Smart Community Security** 07/2024 - 08/2024

Research Assistant in Prof. Weichao Chen's Lab, Tongji University

Aim: Developing an intelligent cross-modal video retrieval system for smart community security surveillance using CLIP-TBPS models

- **Designed multi-modal input processing** supporting text, voice, and image queries for comprehensive surveillance video search capabilities in smart community scenarios
- **Built comprehensive web application** using Flask backend with RESTful APIs, integrating PyTorch for deep learning inference, OpenCV for video processing, and real-time similarity computation
- **Created intelligent result presentation system** displaying top-5 most relevant video clips with precise timestamp annotations on progress bars for efficient content navigation

Github:<https://github.com/Augensternst/TBPS-CLIP>

PROFESSIONAL EXPERIENCE

- **Software Development & Testing Intern, Nanjing Rtunit Technology Co., Ltd.** 07/2024 - 08/2024
 - Participated in software development projects focusing on power electronics and motor drive control systems;
 - Conducted comprehensive software testing including functional testing, performance testing, and system integration testing;
 - Assisted in debugging and optimizing control algorithms for motor drive applications;

AWARDS & HONORS

- Gold Award in the China International College Students' Innovation and Entrepreneurship Competition 2025 Shanghai Regional
- Second Prize in the Semi-finals of the 2nd "SSE Cup" "Hai Ju Ying Cai" Shanghai University Students Innovation and Entrepreneurship Competition in May 2025
- First Prize in the 10th "Excellence Cup" and 19th "Challenge Cup" National University Students' Extracurricular Academic Science and Technology Works Competition at Tongji University in March 2025
- Tongji University Excellent Student Scholarship in 2023 and 2024
- H Prize in the Mathematical Contest in Modeling (MCM), USA in February 2024

SKILLS

- **Languages:** Mandarin (native), English (proficient)
- **Technical Skills**
 - Proficient in using the large model(OpenAI, Qwen,Claude)
 - Computer Language: C/C++, Java, Python, C#, HTML, CSS, JavaScript
 - Database: Oracle, MySQL
 - System: Linux virtual machine
 - Frame:unreal engine, opencv